Algebra 9 — Simultaneous equations (one linear, one non-linear) · Proof

Calculator allowed. Marks in brackets. ★ = harder question. Always substitute the linear equation into the non-linear one.

1 Solve the simultaneous equations $y = x^2 - 3x + 1$ and $y = x + 2$, giving your answers to 3 s.f. [4]

2 Solve the simultaneous equations $x - y = 2$ and $x^2 + y^2 = 20$. [4]

3 Give a counter-example to show that $(x - 3)^2 = x^2 - 9$ is false. [2]

4 Prove that if the difference between two whole numbers is 3, the difference between their squares is a multiple of 3. [2]

5 Prove that the sum of any two consecutive odd numbers is a multiple of 4. [4]

6 **a** Prove that $x^2 - 6x + 10 > 0$ for all values of x. **b** Find the minimum point of the graph of $y = x^2 - 6x + 10$. **c** Sketch the graph. [9]

Total: 25 marks

Check: substitute the linear into the quadratic/circle, then solve the quadratic — each x gives its own y · a counter-example is a single specific case that fails · proofs use n, $2n$ (even), $2n + 1$ (odd), consecutive = $n, n + 1$ · "always positive": complete the square and argue a square is ≥ 0.